

Jupiter on Other Planets

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9:40 PM

let r_{sun} be the distance of a body from the sun -

let a_{sun} be the acceleration a body feels from the sun

let r_{jup} be the distance of a body from jupiter

let a_{jup} be the acceleration a body feels from jupiter.

let p_{body} be the position of body "body".

$$r_{\text{sun}} = p_{\text{sun}} - p_{\text{body}}$$

$$a_{\text{sun}} = \frac{Gm_{\text{sun}}}{r_{\text{sun}}^2} \cdot \frac{r_{\text{sun}}}{|r_{\text{sun}}|}$$

$$r_{\text{jup}} = p_{\text{jup}} - p_{\text{body}}$$

$$a_{\text{jup}} = \frac{Gm_{\text{jup}}}{r_{\text{jup}}^2} \cdot \frac{r_{\text{jup}}}{|r_{\text{jup}}|}$$

$$a_{\text{total}} = a_{\text{sun}} + a_{\text{jup}}$$